

Understanding Analysis, 2nd Edition

Chapter 6 - Section 6.2 - Uniform Convergence of a Sequence of Functions

1. Let

$$f_n(x) = \frac{nx}{1 + nx^2}$$

- (a) Find the pointwise limit of (f_n) for all $x \in (0, \infty)$.
- (b) Is the convergence uniform on $(0, \infty)$?
- (c) Is the convergence uniform on $(0, 1)$?
- (d) Is the convergence uniform on $(1, \infty)$?

Solution:

- (a) Intuitively, as n grows the $+1$ term starts to not matter, so it should converge to $\frac{nx}{nx^2} = \frac{1}{x}$. Formally

$$\lim_{n \rightarrow \infty} \frac{nx}{1 + nx^2} = \lim_{n \rightarrow \infty} \frac{x}{1/n + x^2} = 1/x$$

- (b) No. Given $\epsilon > 0$, let N such that for all $n \geq N$, $|f_n(x) - \frac{1}{x}| < \epsilon$. Or in other words

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} < \epsilon$$

For this to be uniformly convergent, the N needs to work for all x . But the lhs is unbounded as $x \rightarrow 0$; there's always a small enough x such that the lhs is eventually greater than ϵ .

- (c) No, for the same reason as part (b). We can always choose small enough x to break the uniform convergence.
- (d) Now the answer is yes. Reusing the result from part (b), for $x > 1$, the value is upper-bounded by $x = 1$

$$\frac{1}{x(1 + nx^2)} < \frac{1}{1 + n}$$

This doesn't depend on x , so we can simply choose $N \geq \frac{1}{\epsilon} - 1$.

2. (a) Define a sequence of functions on $\mathbf{R}$ by

$$f_n(x) = \begin{cases} 1 & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

and let f be the pointwise limit of f_n .

Is each f_n continuous at zero? Does $f_n \rightarrow f$ uniformly on $\mathbf{R}$? Is f continuous at zero?

- (b) Repeat this exercise using the sequence of functions

$$g_n(x) = \begin{cases} x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

- (c) Repeat once more with

$$h_n(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \\ x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1} \\ 0 & \text{otherwise} \end{cases}$$

In each case, explain how the results are consistent with the Continuous Limit Theorem (Theorem 6.2.6).

Solution:

- (a) Yes, each f_n is continuous at zero. Simply choose $\delta < \frac{1}{n}$, then $f(x) = 0$ for $x < \delta$.

To answer uniform convergence, let's first define f . For x not of the form $\frac{1}{n}$, $f(x)$ is clearly 0. For x of the form $\frac{1}{n}$, then for all $m \geq n$, $f_m(x) = 1$ i.e. eventually all $f_n(x)$ becomes 1. So f is

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

It's not uniformly convergent. Let $\epsilon < 1$. Now say we choose $N = 5$, then for $x = \frac{1}{6}$, we have $f_5(\frac{1}{6}) = 0$ but $f(\frac{1}{6}) = 1$. In general, no single N can work for all x because we can choose $x = \frac{1}{N+1}$, then $|f_N(x) - f(x)| = 1 > \epsilon$.

Finally, f is not continuous at zero. We have $f(0) = 0$. But since $\frac{1}{n} \rightarrow 0$, there's always x of the form $\frac{1}{n}$ close enough to zero such that $f(x) = 1$.

- (b) Each g_n is continuous at zero. Same argument as part (a), choose $\delta < \frac{1}{n}$ then $|x| < \delta \implies g_n(x) = 0$.

Using a very similar argument as part (a), the limit function g is

$$g(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

But unlike part(a), this is uniformly convergent. The idea is to choose N big enough such that even if $g_N(x) = 0$, then x needs to be really small, hence $|g_N(x) - g(x)| = |0 - x| = |x|$ is really small as well.

Formally, choose $N > \frac{1}{\epsilon}$. Then for $n \geq N$, either $g_n(x) = g(x)$ or if $g_n(x) = 0$, then $x = \frac{1}{m}$ with $m > n$. But since $n > \frac{1}{\epsilon}$, if $m > n$ then $\frac{1}{m} < \epsilon$, hence $|g_n(\frac{1}{m}) - g(\frac{1}{m})| = \frac{1}{m} < \epsilon$.

g is continuous at zero, as $\frac{1}{n}$ converges to zero.

- (c) Each h_n is continuous at zero. Choose $\delta < \frac{1}{n}$, then $|x| < \delta \implies h_n(x) = 0$.

Using a similar argument as part (a), for each x of the form $\frac{1}{n}$, we have $h_m(x) = x = \frac{1}{n}$ for all $m > n$. So h is

$$h(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

h is the same as g from part(b), but in this case it's not uniformly convergent. The issue here is that

$$h_N(\frac{1}{N}) - h(\frac{1}{N}) = 1 - \frac{1}{N}$$

So as $x = \frac{1}{n}$ gets smaller, we need to choose increasingly larger N .

More formally, let $\epsilon < \frac{1}{2}$ and $x = \frac{1}{2}$. Choose $N \geq 3$. Then for $n \geq N$, we have

$$h_n(\frac{1}{2}) - h(\frac{1}{2}) = \frac{1}{2} - \frac{1}{2} = 0$$

But this same N cannot for $x = \frac{1}{N+1}$ because for $n = N + 1$ we have

$$h_{N+1}(\frac{1}{N+1}) - h(\frac{1}{N+1}) = 1 - \frac{1}{N+1}$$

And since $N \geq 3$, then $\frac{1}{N+1} \leq \frac{1}{4}$, hence

$$1 - \frac{1}{N+1} \geq 1 - \frac{1}{4} > \epsilon$$

Finally, h is continuous at zero as it's the same as g in part(b).

3. For each $n \in \mathbf{N}$ and $x \in [0, \infty)$, let

$$g_n(x) = \frac{x}{1 + x^n}$$

and

$$h_n(x) = \begin{cases} 1 & \text{if } x \geq 1/n \\ nx & \text{if } 0 \leq x < 1/n \end{cases}$$

- (a) Find the pointwise limit on $[0, \infty)$.
- (b) Explain how we know that the convergence *cannot* be uniform on $[0, \infty)$.
- (c) Choose a smaller set over which the convergence is uniform and supply a proof.

Solution:

(a)

$$g(x) = \begin{cases} x & \text{if } x < 1 \\ 1/2 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

For h , note that for $x > 0$ and large enough n then $x \geq 1/n$ eventually and we always enter the first case.

For $x = 0$, we always enter the second case. So h is

$$h(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(b) For h_n , let's try $\epsilon = 1/4$ and $x = 1/2$ and see what we get.

$h_1(1/2) = \frac{1}{2}$ and $|h_1(1/2) - h(1/2)| = |1/2 - 1| = 1/2 > \epsilon$, so $N = 1$ doesn't work.

$h_2(1/2) = 1$. So $N \geq 2$ works because $h_N(1/2) = 1$.

But this N doesn't work for $x = \frac{1}{N^2}$, because $h_N(\frac{1}{N^2}) = \frac{1}{N}$ and even if we choose the minimum $N = 2$, we get $|\frac{1}{2} - 1| = \frac{1}{2} > \epsilon$. Choosing larger N will only increase the distance from 1 even more.

– **Better solution by chatgpt** –

Use the Continuous Limit Theorem. g_n and h_n is continuous for all n , so if they're uniformly convergent then g and h must be continuous. But g is discontinuous at $x = 1$ and h at $x = 0$.

- (c) For g_n , a simple option is to restrict to $x = 1$. Then $g_n(1) = g(1) = 1/2$ for all n which is certainly uniformly convergent. Likewise for h_n , restrict to $x = 0$. Then $h_n(0) = h(0) = 0$ for all n . But such trivial sets are probably not what the author intended. Let's choose a bigger set.

– **Incorrect** –

For g_n , we can stay away from the discontinuity at 1. Say, $(1, \infty)$. Then $g(x) = 0$ and

$$|g_n(x) - g(x)| = \frac{x}{1+x^n} \leq \frac{x}{x^n} = \frac{1}{x^{n-1}}$$

– **Corrected** –

Oops, $(1, \infty)$ doesn't work because x can still get arbitrarily close to 1. For any N , we can choose x close enough to 1 such that $\frac{1}{x^{n-1}}$ gets close to 1 hence far from $g(x) = 0$.

Instead, we need an interval bounded away from 1, say $[2, \infty)$. then

$$|g_n(x) - g(x)| \leq \frac{1}{x^{n-1}} \leq \frac{1}{2^{n-1}} \rightarrow 0$$

So for any ϵ , choose N big enough so that $\frac{1}{2^{N-1}} < \epsilon$.

h_n is uniformly convergent on $[1, \infty)$ because for $x \geq 1$ we'll always enter the first case, hence $h_n(x) = 1$ for all n and x .

4. TODO. It depends on Exercise 5.2.8 which I haven't done yet.
5. Using the Cauchy Criterion for convergent sequences of real numbers (Theorem 2.6.4), supply a proof for Theorem 6.2.5. (First, define a candidate for $f(x)$, and then argue that $f_n \rightarrow f$ uniformly).

Solution: The proof strategy is very similar to Theorem 2.6.4.

Necessity: uniform convergence $\implies$ Cauchy

There exists N such that for all $n \geq N$, $|f_n(x) - f(x)| < \epsilon/2$ for all x . Then for any $m, n \geq N$ and any x we have

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \epsilon/2 + \epsilon/2 = \epsilon$$

Satisfying Cauchy criterion.

Sufficiency: Cauchy $\implies$ uniform convergence

The Cauchy criterion doesn't say what the limit function is, so we need to define it

first. For a **particular** x , the sequence of real numbers $f_1(x), f_2(x), \dots$ is a Cauchy sequence, so it converges to a limit L_x . Define f to be

$$f(x) = L_x$$

Now we prove uniform convergence. By the Cauchy hypothesis (for sequence of functions), there exists N such that for all $n, m \geq N$, we have

$$|f_n(x) - f_m(x)| < \epsilon/2$$

for any x .

Also, by the fact that $f_1(x), f_2(x), \dots \rightarrow L_x$, there exists a $k \geq N$ such that $|f_k(x) - L_x| < \epsilon/2$. The value of k depends on x , but that's okay because for any x , $k \geq N$ and N does not depend on x .

Combining these two inequalities, we satisfy the condition for uniform convergence. For any $n \geq N$, we have

$$|f_n(x) - L_x| \leq |f_n(x) - f_k(x)| + |f_k(x) - L_x| < \epsilon/2 + \epsilon/2 = \epsilon$$

for any x .

6. TODO

7. Let f be uniformly continuous on all of $\mathbf{R}$, and define a sequence of functions by $f_n(x) = f(x + \frac{1}{n})$. Show that $f_n \rightarrow f$ uniformly. Give an example to show that this proposition fails if f is only assumed to be continuous and not uniformly continuous on $\mathbf{R}$.

Solution: We need to show there exists N such that for all $n \geq N$ and for all x

$$\left| f\left(x + \frac{1}{n}\right) - f(x) \right| < \epsilon$$

By uniform continuity, there exists $\delta > 0$ such that for all x , if $x - \delta < c < x + \delta$ then $|f(c) - f(x)| < \epsilon$. The idea is we just have to pick N (for the uniform convergence) large enough such that $\frac{1}{N} < \delta$.

Since $(\frac{1}{n}) \rightarrow 0$, choose N large enough such that $\frac{1}{N} < \delta$. Then for all $n \geq N$

$$x - \delta < x + \frac{1}{n} < x + \delta \implies \left| f\left(x + \frac{1}{n}\right) - f(x) \right| < \epsilon$$

For the counterexample, take $f(x) = x^2$. This is continuous but not uniformly continuous on $\mathbf{R}$. Then

$$\left| f\left(x + \frac{1}{n}\right) - f(x) \right| = \left| \frac{1}{n^2} + \frac{2x}{n} \right|$$

Suppose there's an N that works for a particular x_1 and ϵ . There exists another x_2 large enough such that $\frac{2x}{n}$ is eventually larger than ϵ , so the convergence is not uniform. More formally, let $\epsilon = 1$. If we choose $x = N$ then

$$\left| \frac{1}{n^2} + \frac{2x}{n} \right| = 2 + \frac{1}{n^2} > 1$$

As to how I knew that x^2 would work as a counterexample, I don't. I just chose the simplest non uniformly continuous function I can think of and it luckily worked. Chatgpt told me the intuition behind this is that " x^2 changes faster if x is large, so shifting by $1/n$ may be tiny, but at large x the vertical change can still be large".

8. Let (g_n) be a sequence of continuous functions that converges uniformly to g on a compact set K . If $g(x) \neq 0$ on K , show $(1/g_n)$ converges uniformly on K to $1/g$.

Solution: I missed the fact that we can use compactness to bound $g(K)$ and only realized it with chatgpt's assistance.

Given $\epsilon > 0$, we need to show that there exists N such that for all $n \geq N$ and for all $x \in K$

$$\left| \frac{1}{g_n(x)} - \frac{1}{g(x)} \right| = \frac{|g_n(x) - g(x)|}{|g_n(x)||g(x)|} < \epsilon$$

The difficulty lies in bounding the denominator. We use the fact that g is continuous on a compact set. By Theorem 4.4.1, $g(K)$ is compact, so it's bounded. And since $0 \notin g(K)$, then there exists a positive lower bound m such that $|g(x)| \geq m$, then

$$\frac{|g_n(x) - g(x)|}{|g_n(x)||g(x)|} \leq \frac{|g_n(x) - g(x)|}{|g_n(x)|m}$$

Since (g_n) converges uniformly to g , we can choose N_1 such that $g_n(x)$ is close to $g(x)$ or equivalently, $|g_n(x)|$ close to $|g(x)|$. Formally, this follows from the fact that

$$||g_n(x)| - |g(x)|| \leq |g_n(x) - g(x)|$$

Choose N_1 such that $|g_n(x)| \geq \frac{m}{2}$, resulting in

$$\frac{|g_n(x) - g(x)|}{|g_n(x)|m} \leq \frac{2}{m^2}|g_n(x) - g(x)|$$

Finally, choose N_2 such that $|g_n(x) - g(x)| < \frac{\epsilon m^2}{2}$ and choose $N = \max(N_1, N_2)$

$$\frac{2}{m^2}|g_n(x) - g(x)| < \frac{2}{m^2} \frac{\epsilon m^2}{2} = \epsilon$$

9. Assume (f_n) and (g_n) are uniformly convergent sequences of functions.

- (a) Show that $(f_n + g_n)$ is a uniformly convergent sequence of functions.
- (b) Give an example to show that the product $(f_n g_n)$ may not converge uniformly.
- (c) Prove that if there exists an $M > 0$ such that $|f_n| \leq M$ and $|g_n| \leq M$ for all $n \in \mathbf{N}$, then $(f_n g_n)$ does converge uniformly.

Solution:

- (a) Let $(f_n) \rightarrow f$ and $(g_n) \rightarrow g$ uniformly, then $(f_n + g_n) \rightarrow (f + g)$.

Choose N_f such that $|f_n(x) - f(x)| < \epsilon/2$ for all x . Choose N_g similarly, and let $N = \max(N_f, N_g)$. Then

$$|(f_n(x) + g_n(x)) - (f(x) + g(x))| \leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \epsilon/2 + \epsilon/2 = \epsilon$$

- (b) Let $f(x) = x$ and $g(x) = \frac{1}{n}$. $f_n \rightarrow f(x) = x$ uniformly, while $g_n \rightarrow g(x) = 0$ uniformly. But $(f_n g_n) = \frac{x}{n}$ does not converge uniformly to $(fg) = 0$ because there exists large enough x so that $\frac{x}{n}$ is big.

Concretely, let $\epsilon = 1/2$. For an N that works for a particular x_1 , the same N does not work for $x_2 = N$ because $\frac{N}{N} = 1 > \epsilon$.

- (c) First, note that if $|f_n| \leq M$ then $|f| \leq M$ as well by Squeeze (if $|f| > M$, then we can make $|f_n(x)|$ within $M + (|f| - M)/2$ of f , a contradiction). The rest follows the typical proof for proving convergence of products

$$\begin{aligned} |f_n(x)g_n(x) - f(x)g(x)| &= |f_n(x)g_n(x) - f(x)g_n(x) + f(x)g_n(x) - f(x)g(x)| \\ &\leq |g_n(x)||f_n(x) - f(x)| + |f(x)||g_n(x) - g(x)| \\ &\leq M|f_n(x) - f(x)| + M|g_n(x) - g(x)| \end{aligned}$$

Then choose N such that $|f_n(x) - f(x)|$ and $|g_n(x) - g(x)|$ are both less than $\frac{\epsilon}{2M}$.

10. Assume $f_n \rightarrow f$ pointwise on $[a, b]$ and the limit function f is continuous on $[a, b]$. If each f_n is increasing (but not necessarily) continuous, show $f_n \rightarrow f$ uniformly.

Solution: This was tough. The following solution is based on this MSE answer.

f is continuous on a compact set so by Theorem 4.4.7 f is uniformly continuous and there exists δ such that $|x - y| < \delta \implies |f(x) - f(y)| < \epsilon/2$. Now, since $[a, b]$ is compact, we can choose a partition $t_0 = a < t_1 < t_2 < \dots < t_k = b$ with $t_i - t_{i-1} < \delta$.

How does this help? Well, the problem with pointwise convergent is that each x may require a different N and since there are infinitely many x , we can't choose a maximum N . The partition helps us to reduce the choice to a *finite* number of N 's.

By pointwise convergence, for each t_i there exists N_i such that $|f_n(t_i) - f(t_i)| < \epsilon/2$. Now we're going to group the choice of an arbitrary t by proving that since $f_n(t_{i-1}) \leq f_n(t) \leq f_n(t_i)$ for some i , then N_i works for t . Then we can pick $N = \max(N_0, N_1, \dots, N_k)$.

We want to show that if $n \geq N$ then $f_n(t) \rightarrow f(t)$ i.e.

$$|f_n(t) - f(t)| < \epsilon$$

By the monotonicity assumption of the problem, we have

$$f_n(t_{i-1}) \leq f_n(t) \leq f_n(t_i)$$

By pointwise convergence on t_i , we have $f_n(t_i) < f(t_i) + \epsilon/2$. Similarly, $f_n(t_{i-1}) > f(t_{i-1}) - \epsilon/2$. Substituting this to the previous inequality, we get

$$f(t_{i-1}) - \epsilon/2 < f_n(t) < f(t_i) + \epsilon/2$$

Now, by uniform continuity of f , we have

$$|f(t) - f(t_i)| < \epsilon/2 \implies f(t_i) < f(t) + \epsilon/2 \implies f(t_i) + \epsilon/2 < f(t) + \epsilon$$

We can similarly prove that $f(t_{i-1}) - \epsilon/2 > f(t) - \epsilon$. Then

$$f(t) - \epsilon < f_n(t) < f(t) + \epsilon$$

As required.

11. TODO

12. TODO

13. The Bolzano-Weierstrass Theorem (BWT) is not true in general for sequence of functions, but under stronger hypotheses several different conclusions are possible. One avenue is to assume the common domain for all of the functions in the sequence is countable. (Another is explored in the next two exercises.)

Let $A = \{x_1, x_2, x_3, \dots\}$ be a countable set. For each $n \in \mathbf{N}$, let f_n be defined on A and assume there exists $M > 0$ such that $|f_n(x)| \leq M$ for all $n \in \mathbf{N}$ and $x \in A$. Follow these steps to show that there exists a subsequence of (f_n) that converges pointwise on A .

- (a) Why does the sequence of real numbers $f_n(x_1)$ necessarily contain a convergent subsequence of (f_n) ? To indicate that the subsequence of functions (f_{n_k}) is generated by considering the values of the functions at x_1 , we will use the notation $f_{n_k} = f_{1,k}$.
- (b) Now, explain why the sequence $f_{1,k}(x_2)$ contains a convergent subsequence.
- (c) Carefully construct a nested family of subsequences $(f_{m,k})$, and show how this can be used to produce a single subsequence of (f_n) that converges at every point of A .

Solution:

- (a) Because by the problem's assumption, $f_n(x_1)$ is bounded by M , so it's a bounded sequence of reals. By BWT, it contains a convergent subsequence. The functions that are part of this subsequence forms $f_{1,k}$.
- (b) $f_{1,k}(x_2)$ is another bounded sequence of reals, so we can apply BWT again to produce $f_{2,k}$.
- (c) The important point of part (b) is that $f_{2,k}$ is a (nested) subsequence of $f_{1,k}$. Since $f_{1,k}$ converges at x_1 , $f_{2,k}$ converges at x_1 as well, in addition to x_2 .

Formally, since A is countable, use induction. The base case is $f_{1,k}$. For the inductive step, let $f_{n,k}$ be a sequence of functions that converges pointwise at $\{x_1, x_2, \dots, x_n\}$. $f_{n,k}(x_{n+1})$ is a bounded sequence of reals, by BWT it contains a convergent subsequence. The functions that are part of the subsequence forms $f_{n+1,k}$ and it converges pointwise on $\{x_1, x_2, \dots, x_n, x_{n+1}\}$.

– **Completion by chatgpt** –

I thought the answer was complete by this point, but chatgpt reminded me that I only have nested subsequences. It remains to produce the single, final subsequence of (f_n) .

Take the diagonal terms $h_k = f_{k,k}$. That is, take the first function of $f_{1,k}$, the second function of $f_{2,k}$, and so on.

Now, for any fixed m , h_k converges on x_m because once $k \geq m$, the tail of h_k is a subsequence of $f_{m,k}$.

14. A sequence of functions (f_n) defined on a set $E \subseteq \mathbf{R}$ is called *equicontinuous* if for every $\epsilon > 0$ there exists a $\delta > 0$ such that $|f_n(x) - f_n(y)| < \epsilon$ for all $n \in \mathbf{N}$ and $|x - y| < \delta$ in E .
- (a) What is the difference between saying that a sequence of functions (f_n) is equicontinuous and just asserting that each f_n in the sequence is individually uniformly continuous?
- (b) Give a qualitative explanation for why the sequence $g_n(x) = x^n$ is not equicontinuous on $[0, 1]$. Is each g_n uniformly continuous on $[0, 1]$?

Solution:

- (a) The difference is that a single δ works for all f_n at once. If each f_n is uniformly continuous but not equicontinuous, then each f_n may require different δ .
- (b) Take $\epsilon = 1/2$. Given any $\delta > 0$, choose $x < 1$ with $|1 - x| < \delta$. Since $x^n \rightarrow 0$ for $0 \leq x < 1$, there exists big enough n such that $|1 - x^n| \geq 1/2$.

Each g_n is uniformly continuous on $[0, 1]$. We have $|x^n - y^n| = |x - y||x^n + x^{n-1}y + \dots + y^n|$. There are n terms in the binomial and the whole expression is bounded by $x = y = 1$, so we have $|x - y||x^{n-1} + x^{n-2}y + \dots + y^{n-1}| \leq n|x - y|$. Pick $\delta < \frac{\epsilon}{n}$.

Or more compactly (pun intended), use Theorem 4.4.7: a function that is continuous on a compact set K is uniformly continuous on K .